

Bueno: Unlocking the operational performance of buildings through smart analytics and AI

Tell us your challenge. We're here to help.

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About Bueno

Bueno helps businesses
operate in a more
sustainable way by
using BigQuery to collate
data sources and
efficient solutions
reducing its costs with Google
Kubernetes Engine

Based in Australia, Bueno is a software as a service (SaaS) company that helps businesses meet their sustainability goals by improving building systems. It uses cloud technology to analyze and collate data and applies algorithms that provide more efficient solutions. With a presence in six countries around the world, Bueno aims to create better places for people to live, work, and play.

Industries: Technology

Google Cloud results

- Reduces the need for certain employees to work after hours, thanks to the stability of Cloud SQL
- Reduces server cost for main database by 20% with the autoscaling feature of Google Kubernetes Engine
- Enables employees to conveniently access and share data from BigQuery using

Location: Australia

BigQu

save c
cost b

Kubernetes Engine

Google Cloud (<https://cloud.google.com/>)

Google Workspace

BigQuery (<https://cloud.google.com/bigquery/>)

Meet

Cloud SQL (<https://cloud.google.com/sql/>) meet/

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Dataflow (<https://cloud.google.com/dataflow/>)^{4/)}

STREETS

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Google Workspace

(<https://workspace.google.com/>)

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Google Meet

(<https://workspace.google.com/products/meet/>)

Contact us (/contact)

Google Chat

(<https://workspace.google.com/products/chat/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

Google Sheets

Buildings are usually equipped with various networks that control and operate the facilities they contain. But air conditioning, lighting, and security systems often exist in their own silos, with no link enabling communication between them. Bueno (<https://www.buenosystems.com.au/>) aims to bridge this gap by using technology to gather data and provide better transparency between the different systems, so that the buildings sector can use this information for fault detection, optimization, and business intelligence.

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*—Scott Horsnell, Chief
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With new technologies being introduced in recent years, there has never been a better time to take advantage of the huge amount of valuable data captured within buildings. "Customers today are looking for a system that is scalable and able to take on a multitude of different datasets," says Scott Horsnell, Chief Innovation Officer at Bueno. "It needs to provide an overview for better context, instead of having various systems working in silos. That's where being in the cloud is beneficial."

A company that was born in the cloud, Bueno has always believed in the concept of scalability. Scott adds that many customers have also approached Bueno for advice about making the move from an on-premises infrastructure to the cloud. "We quite often get asked for assistance in helping them make their first journey into the cloud for operational systems. Our customers look at our solution design

as a method for them to start their own journey into this area."

In order to improve its capability to develop solutions that respond to industry issues relating to data ingestion, structuring, and analysis, Bueno turned to Google Cloud (<https://cloud.google.com/>), because of the ease of use of its managed services.

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manually."

*—Richard Harrison, Chief
Technology Officer, Bueno*

Saving costs and streamlining databases with Google Kubernetes Engine

Because Bueno uses a microservices-based architecture, as the company grew, it wanted to switch to using Kubernetes as its orchestration solution, so that it could manage the growing complexities without spending too much time on operations. Without much experience in running Kubernetes, Bueno chose Google Kubernetes Engine (/kubernetes-engine) (GKE) because of its ease of use.

"This was our first deployment of Kubernetes, and we thought that it would be far better to have something that was managed, rather than building and managing it ourselves," says Richard.

"Furthermore, the autoscaling and auto-upgrade abilities of GKE means that the team can spend more time working on the core business rather than fixing operational issues."

Prior to GKE, Bueno had to manually ensure that it had enough processing capacity for a variety of scheduled jobs. After deploying GKE, it has seen its infrastructure costs reduce significantly. "Since moving to Google Kubernetes Engine, we rely on the autoscaling feature to give us compute power only

when we need it," says Richard. This has saved Bueno significant cost because a lot of the infrastructure management work before this had to be done manually.



Maximizing the potential of data with BigQuery

One of the goals for Bueno when migrating to Google Cloud was to use existing data to better understand its customers and the business. Tapping into the capabilities of [Pub/Sub \(/pubsub\)](#) and [Dataflow \(/dataflow\)](#) to ingest data from a range of different systems, Richard says, "We've been able to move all of our major data sources into [BigQuery \(/bigquery\)](#), which helps us understand the different aspects of our customers such as equipment and site level data, weather data, and even the last visits," says Richard.

Customers can also use Bueno's system to look for insights and discover new things that may require action being taken. "From the data we provide, customers can generate a work order for maintenance to be done and log the cost of the job,"

he adds. "On our end, it means that we can use that data to verify the value and accuracy of our analytics."

The machine learning capabilities of BigQuery also mean that Bueno can deliver enhanced preventative maintenance, so that customers can diagnose and fix a product before it breaks down at an earlier stage. Currently, most of Bueno's customers perform routine manual checks, which can be very time consuming and take up a lot of work hours. "With the help of BigQuery, we have been able to detect problems earlier and ensure that their equipment operates at optimum capabilities. This has saved as much as 60% of our customer's cost," says Scott.

A smooth onboarding process

Even though Bueno is new to Google Cloud, with the range of online courses available, it didn't take long for the team to familiarize itself with the tools.

Equipped with a comprehensive list of courses, the team could easily learn everything they needed to know prior to the actual migration. "One of the first things Google Cloud did was to give us access to online courses and tutorials, so that our engineering team could prepare themselves before making the move," says Richard. "With all the resources online, we could easily get up to speed. This made the migration process seamless, but more than that, it became a great learning and development opportunity for employees."

According to Richard, the majority of its core database workload was migrated within just a few

weeks of making the decision to switch to Google Cloud, with no disruption to the workflow. "We've had some fantastic results," he says. "Some early results show that we've experienced a 20% reduction in cost for our main database and analytical workload."

There has also been a positive impact on the daily workload of its employees. "We have found Cloud SQL (/sql) to be really stable, and we haven't experienced a single outage," says Richard. "Our employees now have a better work-life balance, as they now rarely have to work after hours due to the stable performance of Cloud SQL."

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*—Richard Harrison, Chief
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Seamless connectivity with Google Workspace

Even though it recently migrated to Google Cloud to manage its customer data, Bueno has been a Google Workspace (<https://workspace.google.com/>) customer since the company was founded. "Being on Google Workspace means we can use the same sign-on to access our cloud environment," says Richard. "This makes accessibility and user management much easier, and was one of the things that attracted us to Google Cloud in the first place."

The team has already been using products such as Google Meet (<https://workspace.google.com/products/meet/>) and Google Chat (<https://workspace.google.com/products/chat/>) in their day-to-day operations, but what stands out for them, apart from platform familiarity, is the seamless integration between Google Cloud products and

Google Workspace. "We really like the Sheets

(<https://workspace.google.com/products/sheets/>)

integration with BigQuery, it's so handy for sharing data," Richard says. "I can write a complicated query and use Sheets to boil it down into a very simple dataset for my colleagues who would use that data to make smarter decisions for the company. They can also update that data and immediately share it with people anywhere around the world."

Beyond the products, Scott shares that his experience working with Google Cloud has been a pleasant one. "The relationship that we've built with our Google Cloud representative has been a real highlight," he says. "He is always quick to respond to our questions and provides us access to relevant experts when we need it. It has been a refreshing experience and unlike any other vendor we've worked with."